

1. Aufgabe

a) $i(t) = I_{\max}$ für $0 \leq t \leq T/3$

$$i(t) = 2 I_{\max} - \frac{2 I_{\max}}{\frac{2}{3} T} \cdot t$$

$$= 2 I_{\max} - \frac{3 I_{\max}}{T} \cdot t \quad \text{für } \frac{T}{3} \leq t \leq \frac{2}{3} T$$

$$i(t) = 2 I_{\max} - \frac{3 I_{\max}}{T} \cdot t \quad \text{für } \frac{T}{3} \leq t \leq \frac{2}{3} T$$

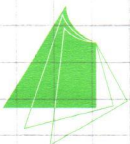
(ist ja die gleiche Funktion !)

b) $\bar{i}(t) = \frac{1}{T} \int_0^T i(t) dt = \frac{1}{T} \left(\underbrace{\frac{1}{3} T I_{\max}}_{\substack{\text{Fläche} \\ \text{Rechteck}}} + \underbrace{\text{Dreieck}}_{\substack{\text{Dreiecksflächen} \\ \text{heben sich auf!}}} + \underbrace{\text{Dreieck}}_{\substack{\text{Dreiecksflächen} \\ \text{heben sich auf!}}} \right)$

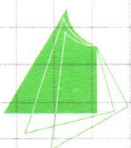
$$\bar{i}(t) = \underline{\underline{\frac{1}{3} I_{\max}}}$$

c) $\bar{|i|} = \frac{1}{T} \int_0^T |i(t)| dt = \frac{1}{T} \left[\frac{1}{3} T \cdot I_{\max} + 2 \cdot \underbrace{\frac{1}{3} T I_{\max} \cdot \frac{1}{2}}_{\substack{\text{Dreieck-Fläche} \\ \text{2mal}}} \right]$

$$= \frac{1}{T} \left[\frac{1}{3} T I_{\max} + \frac{1}{3} T I_{\max} \right]$$
$$= \underline{\underline{\frac{2}{3} I_{\max}}}$$



$$\begin{aligned}
 d) \quad I_{RMS} &= \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} \\
 &= \sqrt{\frac{1}{T} \left[\int_0^{\frac{1}{3}T} I_{max}^2 dt + \int_{\frac{1}{3}T}^T \left(2 I_{max} - \frac{3 I_{max}}{T} t \right)^2 dt \right]} \\
 &= \sqrt{\frac{1}{T} \left[I_{max}^2 \cdot \frac{1}{3}T + \int_{\frac{1}{3}T}^T \left(4 I_{max}^2 - 12 I_{max}^2 \frac{t}{T} + \frac{9 I_{max}^2}{T^2} t^2 \right) dt \right]} \\
 &= I_{max} \sqrt{\frac{1}{T} \left[\frac{1}{3}T + \left(4t - 12 \cdot \frac{t^2}{2T} + \frac{9}{T^2} \cdot \frac{t^3}{3} \right) \Big|_{\frac{1}{3}T}^T \right]} \\
 &= I_{max} \sqrt{\frac{1}{T} \left[\frac{1}{3}T + \left(4 \cdot \frac{2}{3}T - \frac{6}{T} \left(T^2 - \left(\frac{1}{3}T \right)^2 \right) + \frac{3}{T^2} \left(T^3 - \left(\frac{1}{3}T \right)^3 \right) \right) \right]} \\
 &= I_{max} \sqrt{\frac{1}{3} + \frac{8}{3} - \frac{16}{3} + \frac{26}{9}} \\
 &= \underline{\underline{I_{max} \cdot \sqrt{\frac{5}{9}} \approx 0,75 I_{max}}}
 \end{aligned}$$



2. Aufgabe

$$\begin{aligned} 2a) \quad X_{C1} &= \frac{1}{\omega C_1} = \frac{1}{2\pi f \cdot C_1} = \frac{1}{2\pi \cdot 1591,54 \cdot \frac{1}{3} \cdot 33,33 \text{ nF}} \\ &= 3000,31 \Omega \approx \underline{\underline{3 \text{ k}\Omega}} \end{aligned}$$

$$\begin{aligned} X_{L1} &= X_{L2} = \omega L = 2\pi f \cdot L_1 \\ &= 2\pi \cdot 1591,54 \cdot \frac{1}{3} \cdot 0,2 \text{ H} = \underline{\underline{2 \text{ k}\Omega}} \end{aligned}$$

$$b) \quad U_{R2} = 4 \text{ V} \Rightarrow \underline{I_2} = \frac{U_{R2}}{R_2} = \frac{4 \text{ V}}{1 \text{ k}\Omega} = \underline{\underline{4 \text{ mA}}}$$

$$\underline{U_{L2}} = j X_{L2} \cdot \underline{I_2} = j 2 \text{ k}\Omega \cdot 4 \text{ mA} = \underline{\underline{j 8 \text{ V}}}$$

$$\underline{U_{C1}} = -j X_{C1} \cdot \underline{I_2} = -j 3 \text{ k}\Omega \cdot 4 \text{ mA} = \underline{\underline{-j 12 \text{ V}}}$$

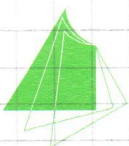
$$\begin{aligned} \underline{U_E} &= \underline{U_{C1}} + \underline{U_{L2}} + \underline{U_{R2}} = -j 12 \text{ V} + j 8 \text{ V} + 4 \text{ V} \\ &= \underline{\underline{4 \text{ V} - j 4 \text{ V}}} = \sqrt{2} \cdot 4 \text{ V} e^{-j 45^\circ} \end{aligned}$$

$$\begin{aligned} \underline{I_1} &= \frac{\underline{U_E}}{R_1 + j X_{L1}} = \frac{\sqrt{2} \cdot 4 \text{ V} \cdot e^{-j 45^\circ}}{1 \text{ k}\Omega + j 2 \text{ k}\Omega} \\ &= \frac{\sqrt{2} \cdot 4 \text{ V} \cdot e^{-j 45^\circ}}{\sqrt{(1 \text{ k}\Omega)^2 + (2 \text{ k}\Omega)^2} e^{j \arctan(2/1)}} = \frac{\sqrt{2} \cdot 4 \text{ V} \cdot e^{-j 45^\circ}}{\sqrt{5} \text{ k}\Omega \cdot e^{j 63,4^\circ}} \\ &= \underline{\underline{2,53 \text{ mA} e^{-j 108,4^\circ}}} = (-0,8 - j 2,4) \text{ mA} \end{aligned}$$

$$\underline{U_{R1}} = R_1 \cdot \underline{I_1} = \underline{\underline{2,53 \text{ V} e^{-j 108,4^\circ}}} = (-0,8 - j 2,4) \text{ V}$$

$$\begin{aligned} \underline{U_{L1}} &= j X_{L1} \cdot \underline{I_1} = j 2 \text{ k}\Omega \cdot (-0,8 - j 2,4) \text{ mA} \\ &= \underline{\underline{4,8 \text{ V} - j 1,6 \text{ V}}} \end{aligned}$$

$$\begin{aligned} \underline{I_E} &= \underline{I_1} + \underline{I_2} = (-0,8 - j 2,4) \text{ mA} + 4 \text{ mA} \\ &= \underline{\underline{3,2 \text{ mA} - j 2,4 \text{ mA}}} = 4 \text{ mA} \cdot e^{-j 36,9^\circ} \end{aligned}$$



$$c) \quad \underline{I}_E = 4 \text{ mA} \cdot e^{-j36,9^\circ}$$

$$\Rightarrow i_E(t) = \sqrt{2} \cdot 4 \text{ mA} \cos(\omega t - 36,9^\circ) = \underline{\underline{5,66 \text{ mA} \cdot \cos(\omega t - 36,9^\circ)}}$$

$$\underline{U}_E = \sqrt{2} \cdot 4 \text{ V} \cdot e^{-j45^\circ}$$

$$\Rightarrow u_E(t) = \underline{\underline{8,0 \text{ V} \cos(\omega t - 45^\circ)}}$$

$$d) \quad \frac{\underline{I}_{E,\text{neu}}}{\underline{I}_E} = \frac{\underline{U}_{E,\text{neu}}}{\underline{U}_E} \quad (\text{Zeigerdiagramm bleibt ähnlich})$$

$$\begin{aligned} \Rightarrow \underline{I}_{E,\text{neu}} &= \frac{\underline{U}_{E,\text{neu}}}{\underline{U}_E} \underline{I}_E \\ &= \frac{10 \text{ V} \cdot e^{j45^\circ}}{\sqrt{2} \cdot 4 \text{ V} \cdot e^{-j45^\circ}} \cdot 4 \text{ mA} \cdot e^{-j36,9^\circ} \\ &= \underline{\underline{7,07 \text{ mA} \cdot e^{j53,1^\circ}}} \end{aligned}$$

$$e) \quad \text{Gleichspannung!} \Rightarrow \omega = 0$$

C sperrt, L Kurzschluss

$$\underline{I}_E = I_E = \frac{U_E}{R_1} = \frac{10 \text{ V}}{1 \text{ k}\Omega} = \underline{\underline{10 \text{ mA}}}$$

f) Stromteilerregel verwenden:

$$\underline{I}_1 = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} \cdot \underline{I}_E = \frac{(-jX_{L1} + jX_{L2} + R_2)}{(R_1 + jX_{L1}) + (R_2 - jX_{L1} + jX_{L2})} \underline{I}_E$$

